LECTURE SERIES · DATABASE SYSTEMS

Types of Databases

From Relational Tables to Data-Driven Decisions

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PostgreSQL · MongoDB · BigQuery · InfluxDB · Analytics

TODAY'S ROADMAP

What We Will Cover

Each database type is built for a different **shape of data** and **workload**.

01
Relational
`PostgreSQL`

02
NoSQL
`MongoDB`

03
Warehouse
`BigQuery`

04
Time-Series
`InfluxDB`

05
Analytics
`Data → Insight`

MOTIVATION

Why Not One Database?

Different workloads need different strengths.

Workload	Needs most	Best fit
Bank transfer	Correctness	PostgreSQL
Product catalog	Flexibility	MongoDB
10 years of sales	Fast analysis	BigQuery
Sensor every second	Fast writes	InfluxDB

TYPE 1 · RELATIONAL DATABASE

PostgreSQL (RDBMS)

Data in **tables of rows and columns**, linked by **keys**, queried with **SQL**.

ACID

Safe, all-or-nothing transactions

Fixed schema

Structure defined first

Joins

Tables linked by foreign keys

Open source

Free and extensible

BEST FOR

Banking, ERP, student records

LIMITATION

Rigid schema; harder to scale out

TYPE 2 · BEYOND TABLES

NoSQL: “Not Only SQL”

Flexible-schema databases built to **scale out** across many servers.

Document

JSON-like records

MongoDB

Key–Value

Fast key lookups

Redis

Wide-Column

Huge write volume

Cassandra

Graph

Nodes & edges

Neo4j

Trade-off: many NoSQL systems relax strict consistency (BASE) to gain speed and availability.

NOSQL DOCUMENT STORE

MongoDB

Each record is a flexible **document** — nested fields and lists in one place.

ONE STUDENT = ONE DOCUMENT

```
{
  "name": "Rahim",
  "dept": "CSE",
  "phones": ["0171…", "0181…"],
  "city": "Saidpur"
}
```

- **Flexible schema**
- **Sharding** — scales out
- **Replica sets** — always on

Best for: profiles, catalogs, apps

TYPE 3 · ANALYTICAL STORAGE

Data Warehouse

A central store of **historical data**, built for **analysis**.

	OLTP	OLAP (warehouse)
Purpose	Run the business	Analyse the business
Work	Small updates	Large summaries
Data	Current	Historical
Example	PostgreSQL	BigQuery

Extract → **Transform** → **Load** → **Reports**

CLOUD DATA WAREHOUSE

Google BigQuery

Fully managed: write SQL, and Google runs it on thousands of machines.

Serverless
No servers to manage

Columnar
Reads only needed columns

Huge scale
Terabytes in seconds

Pay per use
Billed by data scanned

Best for: business reports, dashboards, big-data analytics

TYPE 4 · TIME-SERIES DATABASE

InfluxDB

Time-series data = measurements stamped with time, arriving fast.

Data model

Measurement — what is measured

Tags — labels (room, city)

Fields — values (temp, speed)

Timestamp — when it happened

- Very fast writes
- Time-window queries
- Old data auto-deleted

Best for: IoT, server monitoring

SIDE BY SIDE

Comparing the Four

	PostgreSQL	MongoDB	BigQuery	InfluxDB
Type	Relational	Document	Warehouse	Time-series
Data	Tables	Documents	Columns	Time points
Schema	Strict	Flexible	Defined	Flexible
Workload	Transactions	Web apps	Analytics	Metrics
Example	Banking	Catalog	Reports	Sensors

No single winner — choose by your data and workload.

PART 2 · FROM DATA TO DECISIONS

Four Types of Data Analytics

Data analytics finds patterns in data to support decisions.

1 **Descriptive**
What happened?

2 **Diagnostic**
Why did it happen?

3 **Predictive**
What will happen?

4 **Prescriptive**
What should we do?

THE ANALYTICS LIFECYCLE

Data Analytics Process

1 Ask
Define the question

2 Collect
Gather the data

3 Clean
Fix errors & gaps

4 Analyze
Find patterns

5 Visualize
Charts & dashboards

6 Act
Decide & repeat

Common tools: SQL, Python (pandas), Excel, Power BI, Tableau, Looker Studio

WRAP-UP

Key Takeaways

Remember

- **PostgreSQL** — safe transactions
- **MongoDB** — flexible documents
- **BigQuery** — large-scale analytics
- **InfluxDB** — time-stamped data
- **Analytics** — data to decisions

Review questions

1. Why does a bank need ACID?
2. When is MongoDB better than SQL?
3. Differentiate OLTP and OLAP.
4. Name two uses of time-series data.
5. Name the four types of analytics.